BRAINSTORMING & IDEA PRIORITIZATION

Project: FitBuddy | Track: AI/ML & Generative AI

1. Ideation Scope & Innovation Objectives

FitBuddy is a web-based AI fitness planning application designed to turn basic user fitness information into a structured, personalized 7-day workout plan with nutrition or recovery guidance. The idea combines deterministic user-input validation with Generative AI for flexible plan generation and feedback-based revision.

2. Evaluated Solution Concepts

- Personalized 7-Day Workout Generator: Generates a structured weekly plan from age, weight, goal, and intensity.
- AI Nutrition & Recovery Assistant: Produces concise goal-based hydration, nutrition, and recovery guidance.
- Feedback-Based Plan Updater: Uses the original plan plus user feedback to regenerate a revised plan.
- Admin/Coach Dashboard: Allows review of registered users and original/updated plans.

Candidate Feature Concept	Feasibility	User Impact	GenAI Value	Verdict
Personalized workout generator	5 / 5	5 / 5	5 / 5	Accepted (P0)
Feedback-based plan updater	4 / 5	5 / 5	5 / 5	Accepted (P0)
Nutrition/recovery tips	5 / 5	4 / 5	4 / 5	Accepted (P1)
Admin/coach dashboard	4 / 5	4 / 5	2 / 5	Accepted (P1)

■ Phase 1 Takeaway

- Prioritized the core workout-generation and feedback loop as the main product value.
- Kept the architecture simple enough for reliable local development and testing.

PROJECT SCOPE & REQUIREMENTS

FitBuddy | Functional and Non-Functional Requirements

1. Functional Scope

- Collect user name, user ID, age, weight, fitness goal, and preferred workout intensity.
- Generate a customized 7-day workout plan through Gemini.
- Generate a concise nutrition or recovery tip.
- Accept feedback such as more cardio, more rest days, or yoga and revise the plan.
- Persist users and plan versions in SQLite.
- Provide an admin/coach view of registered users and plans.

2. Non-Functional Requirements

- Simple browser-based interface with responsive layout.
- Fast API response flow suitable for local demonstration.
- Clear separation of frontend, backend, AI, and database responsibilities.
- API key stored outside source code using environment variables.
- Readable, maintainable Python modules and HTML templates.

Requirement	Priority	Validation Method
7-day workout generation	P0	Submit valid user form and inspect result
Feedback-based revision	P0	Submit feedback and compare plan versions
Nutrition/recovery tip	P1	Verify tip is displayed with result
Database persistence	P0	Reload/retrieve stored user and plan
Admin dashboard	P1	Open all-users view and inspect records

■ Phase 2 Takeaway

- Core scope is centered on personalization, generation, revision, and persistence.
- Requirements map directly to demonstrable application pages and API routes.

PROBLEM STATEMENT & OBJECTIVES

FitBuddy | Problem Definition and Project Goals

1. Problem Statement

Many beginners need structured fitness guidance but may find generic routines difficult to personalize. FitBuddy addresses this by converting a small set of user preferences into a structured weekly plan and allowing the user to request changes through natural-language feedback.

2. Project Objectives

- Automate the creation of structured 7-day workout schedules.
- Adapt workout content to the user's selected goal and intensity.
- Provide practical nutrition or recovery guidance alongside the plan.
- Enable iterative improvement through natural-language feedback.
- Store user and plan information for later review.

3. Intended Users

- Individual users seeking a structured starting fitness plan.
- Coaches or administrators reviewing generated plans.
- Students demonstrating a practical Generative AI web application.

Objective	Expected Output	Success Indicator
Personalization	Goal/intensity-aware plan	Plan reflects submitted inputs
AI generation	7-day structured routine	All seven days are represented
Revision	Updated plan	Requested change appears in revision
Persistence	Stored records	Records available after submission

■ Phase 3 Takeaway

- The project solves a clear personalization problem while remaining suitable for an academic AI/ML demonstration.

SOLUTION DESIGN

FitBuddy | End-to-End Functional Design

1. Solution Flow

User input → FastAPI validation → Gemini workout generation + Gemini nutrition/recovery generation → SQLite persistence → Jinja2 result page → user feedback → Gemini plan update → updated plan stored and displayed.

2. Main Application Modules

- routes.py – application routes and orchestration.
- gemini_generator.py – workout-plan generation.
- gemini_flash_generator.py – nutrition/recovery tip generation.
- updated_plan.py – feedback-driven plan revision.
- database.py – SQLAlchemy persistence and retrieval.
- templates – Jinja2 pages for input, results, and administration.

Layer	Responsibility	Main Technology
Presentation	Forms, result display, admin table	HTML/CSS/Jinja2
Application	Routing and validation	FastAPI/Pydantic
AI	Generation and revision	Google Gemini
Persistence	Users and plan versions	SQLite/SQLAlchemy
Runtime	Local ASGI server	Uvicorn

■ Phase 4 Takeaway

- A modular design makes the project easier to debug, demonstrate, and extend.

AI MODEL SELECTION

FitBuddy | Generative AI Evaluation

1. AI Requirements

- Generate structured day-by-day workout content.
- Respond to user goals and intensity settings.
- Revise an existing plan using natural-language feedback.
- Produce concise nutrition or recovery suggestions.
- Integrate with a FastAPI web application.

2. Models Considered

- T5 and BART – text-generation alternatives.
- DistilBERT – possible intent-classification component.
- BLOOM – general-purpose text generation.
- Gemini – selected for API-based integration and structured generation.

Model/Approach	Primary Use	Integration Fit	Decision
T5 / BART	Text generation	Moderate	Not selected
DistilBERT	Intent classification	Moderate	Not required for MVP
BLOOM	General generation	Moderate	Not selected
Gemini	Workout + feedback + tips	High	Selected

■ Phase 5 Takeaway

- Gemini was selected as the documented final approach for practical API integration and structured output.

TECHNICAL ARCHITECTURE

FitBuddy | Frontend, Backend, AI and Database Architecture

1. Architecture Overview

FitBuddy uses a modular FastAPI architecture. The frontend collects input and displays results; the backend validates requests and coordinates services; the AI layer generates or revises content; and SQLite stores user and plan records.

2. Architecture Components

- Frontend: index.html, result.html, all_users.html.
- Backend: FastAPI routes and Pydantic validation.
- AI services: Gemini workout, nutrition, and update functions.
- Database: SQLite with SQLAlchemy ORM.
- Server: Uvicorn ASGI development server.

Component	Input	Output
Frontend	User form / feedback	HTTP form request
FastAPI	Validated request	Service call + template context
Gemini	Prompt + user/plan data	Generated text
SQLite	User and plan records	Persistent data
Jinja2	Template + context	Rendered HTML

■ Phase 6 Takeaway

- The architecture separates concerns and provides a clear path from user interaction to AI-generated output.

DATABASE & DATA MANAGEMENT

FitBuddy | SQLite + SQLAlchemy Persistence

1. Stored Information

- User ID and name.
- Age and weight.
- Fitness goal and preferred intensity.
- Original generated workout plan.
- Updated plan versions created from feedback.

2. Database Operations

- save_user() – store user details.
- save_plan() – store an original generated plan.
- update_plan() – store a revised plan.
- get_original_plan() – retrieve the original plan.
- get_user() / get_all_users() – retrieve user records.
- get_all_plans() – retrieve stored plans.

Entity	Key Data	Purpose
User	ID, name, age, weight, goal, intensity	Identify and personalize
Plan	User ID, plan text, version	Store generated content
Revision	Original/updated content	Preserve feedback history

■ Phase 7 Takeaway

- Separating original and revised plans preserves the feedback loop and supports admin review.

AI FUNCTIONALITIES

FitBuddy | Workout Generation, Tips and Feedback Revision

1. Workout Plan Generation

generate_workout_gemini() receives user fitness inputs and asks Gemini to create a structured 7-day routine. The documented structure includes warm-up, main workout details such as exercises/sets/repetitions, and cooldown or recovery guidance.

2. Nutrition / Recovery Tip

generate_nutrition_tip_with_flash() creates a concise goal-based suggestion, such as hydration, protein, or recovery guidance.

3. Feedback-Based Updating

update_workout_plan() receives the original plan and user feedback, then asks Gemini to produce a revised plan containing the requested changes.

Function	Model Role	Output
generate_workout_gemini()	Structured generation	7-day workout plan
generate_nutrition_tip_with_flash()	Fast guidance	Nutrition/recovery tip
update_workout_plan()	Revision	Updated workout plan

■ Phase 8 Takeaway

• Three focused AI functions keep the Generative AI layer understandable and reusable.

FASTAPI BACKEND & ROUTING

FitBuddy | API Endpoints and Request Processing

1. Core Routes

- / – displays the home page.
- /generate-workout – accepts user details and generates a plan and tip.
- /submit-feedback – retrieves the stored plan and creates an updated version.
- /view-all-users – displays the admin/coach dashboard.

2. Validation and Processing

- FastAPI receives HTML form data using Form(...).
- Pydantic schemas such as UserInput and FeedbackRequest validate structured data.
- Routes call the appropriate AI helper functions.
- Database functions save and retrieve users and plan versions.
- Template responses render the final browser pages.

Route	Request	Main Action	Response
/	GET	Show input page	index.html
/generate-workout	POST	Generate + save plan	result.html
/submit-feedback	POST	Revise + save plan	result.html
/view-all-users	GET	List records	all_users.html

■ Phase 9 Takeaway

- The backend acts as the central coordinator between UI, Gemini services, and SQLite.

FRONTEND DEVELOPMENT

FitBuddy | Responsive Jinja2 Web Interface

1. index.html – User Input

- Collects name, user ID, age, weight, goal, and intensity.
- Uses grouped form fields and clear labels.
- Submits the form to /generate-workout.

2. result.html – AI Output

- Displays the username, goal, intensity, generated workout plan, and nutrition/recovery tip.
- Provides a feedback form for requested changes.
- Uses readable preformatted blocks for the day-by-day plan.

3. all_users.html – Admin View

- Displays registered users and plan versions.
- Supports review of original and updated plans.
- Can be extended with user deletion controls.

UI Feature	Implementation
Theme	Dark fitness-oriented layout
Typography	Roboto / clean sans-serif
Layout	Centered Flexbox + responsive media queries
Controls	Rounded inputs and hover-enabled buttons
Output	Structured result blocks and tables

■ Phase 10 Takeaway

- The UI is designed to keep the user flow simple: enter details → generate → review → give feedback → revise.

PROJECT WORKFLOW & USER JOURNEY

FitBuddy | From User Input to Updated Plan

1. Expected Application Flow

- User opens the FitBuddy home page.
- User enters personal details and selects goal and intensity.
- FastAPI receives and validates the form.
- Gemini generates the structured 7-day workout plan.
- Gemini Flash generates the nutrition/recovery tip.
- The result page displays both outputs.
- User submits optional feedback.
- The original plan plus feedback is sent for revision.
- The updated plan is saved and displayed.
- Admin/coach can review user and plan records.

Stage	System Action	User/Administrator View
1	Collect inputs	User form
2	Generate	7-day plan + tip
3	Review	Result page
4	Feedback	Requested changes
5	Revise	Updated plan
6	Persist	Database record
7	Monitor	Admin dashboard

■ Phase 11 Takeaway

- The feedback loop is the key differentiating workflow because the user can iteratively refine the generated plan.

TESTING & VERIFICATION

FitBuddy | Functional Testing Checklist

1. Test Areas

- Verify that the home page loads without template errors.
- Submit valid user details and confirm a 7-day plan is returned.
- Confirm nutrition/recovery guidance appears with the plan.
- Submit feedback and confirm a revised plan is produced.
- Confirm original and updated plans are preserved in SQLite.
- Verify the admin page lists users and plan information.
- Check responsive behavior on smaller screens.
- Verify missing or invalid input is handled cleanly.

Test Case	Expected Result	Status Target
Home page	Form loads	Pass
Valid generation	7-day plan displayed	Pass
Nutrition tip	Tip displayed	Pass
Feedback	Updated plan displayed	Pass
Persistence	Records retrievable	Pass
Admin view	Users/plans visible	Pass
Invalid input	Clear validation response	Pass

■ Phase 12 Takeaway

- Testing should cover both the normal AI workflow and basic validation/database failure paths before demonstration.

DEPLOYMENT, CONCLUSION & FUTURE SCOPE

FitBuddy | Local Deployment and Project Summary

1. Local Deployment

- Create a virtual environment: python -m venv venv
- Activate it: venv\Scripts\activate on Windows.
- Install dependencies with pip install -r requirements.txt.
- Store the Gemini API key in an environment variable or .env file.
- Run with: uvicorn app.main:app --reload
- Open the application at http://127.0.0.1:8000 and API docs at /docs.

2. Conclusion

FitBuddy combines FastAPI, Gemini, Jinja2, HTML/CSS, SQLite, and SQLAlchemy into a modular Generative AI application for personalized fitness planning. Its documented capabilities include 7-day workout generation, nutrition/recovery guidance, feedback-based revision, database persistence, and administrative review.

3. Future Scope

- Authentication and role-based admin access.
- Exercise image/video support and safer exercise explanations.
- Progress tracking and plan history charts.
- More structured output validation using JSON schemas.
- Cloud database and deployment for multi-user access.
- Additional wellness modules such as sleep and habit tracking.

Area	Current Project	Future Enhancement
AI	Gemini generation/revision	Structured JSON + stronger validation
Database	SQLite	Cloud relational database
Admin	Review users/plans	Authentication + analytics
User Experience	Responsive web UI	Progress dashboard + history

■ Final Takeaway

- FitBuddy provides a complete academic Generative AI workflow from user input to personalized generation, feedback-based revision, storage, and review.
- The modular design leaves clear paths for security, validation, analytics, and deployment improvements.